

PULI SAI TEJA

Software Engineer | Python & Backend APIs | CI/CD & Reliability
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PROFILE

Software Engineer with 2+ years at TCS building scalable backend services, RESTful APIs, and reliable production systems in Linux environments. Proficient in Python, Docker, Kubernetes, and CI/CD, with a strong focus on fault tolerance, secure coding practices, and system observability. Experienced using AI coding tools (GitHub Copilot, Claude) to accelerate development, testing, and code review, plus applied machine learning for automated detection and classification. Maintains production environments, debugs issues, and resolves reliability problems efficiently.

PROFESSIONAL EXPERIENCE

Software Engineer, Backend & Reliability — Tata Consultancy Services Jul 2024 – Present

- Designed and maintained backend services and RESTful APIs in Python (FastAPI, Django REST Framework) on Linux, building secure, performant, and fault-tolerant systems with PostgreSQL, Redis, and MySQL.
- Used AI coding tools (GitHub Copilot, Claude) to accelerate development, unit testing, and code review — reviewing AI-generated code to identify issues and ensure correctness against the codebase.
- Built and maintained CI/CD pipelines (Jenkins, GitHub Actions) and deployed containerised services using Docker and Kubernetes across production clusters at 99.9% uptime SLA.
- Maintained production environments and participated in incident response — debugging software issues, performing root cause analysis, and resolving reliability problems, reducing recurrence of known issues by 45%.
- Prototyped Python + Ansible automation and tooling that improved reliability and operational efficiency, cutting manual effort by ~40% and deployment time from 2 hours to under 30 minutes.
- Instrumented systems for observability with structured logging, monitoring, and alerting (Prometheus, Grafana, ELK), enabling proactive detection and reducing MTTR by 30%.

Intern — ISRO — Indian Space Research Organisation Nov 2022 – Jan 2023

- Contributed to 2 cross-functional R&D projects, delivering assigned modules ahead of schedule and collaborating with senior engineers across design and documentation workflows.

TECHNICAL SKILLS

Languages: Python, Java, C/C++, SQL, Bash; OOP, data structures, secure coding practices

Backend & APIs: FastAPI, Django, Django REST Framework, RESTful API design, microservices, fault-tolerant systems

CI/CD & Containers: Docker, Kubernetes, Jenkins, GitHub Actions, GitLab CI/CD, Git, Ansible

AI & ML: AI coding assistants (GitHub Copilot, Claude), TensorFlow, scikit-learn, CNNs, applied ML for detection/classification

Data & Databases: PostgreSQL, Redis, MySQL, Pandas, NumPy; data pipelines and transformation

Reliability & Cloud: Linux/RHEL, AWS (EC2, S3, RDS), GCP, Nginx, Prometheus, Grafana, ELK, incident response, on-call

PROJECTS

Obscene Content Classification (Applied ML) 2023 — Built a CNN detection model with 91% accuracy on 10,000+ images (TensorFlow, scikit-learn) for automated content moderation; preprocessing pipeline cut training time 25% — an AI-driven detection workflow.

Indoor Navigation System 2024 — Backend REST APIs (Django REST Framework + PostgreSQL) with Dijkstra shortest-path (95% accuracy); full SDLC ownership, deployed on Linux/Nginx.

Car Rental Web Application 2023 — Full-stack Django + PostgreSQL app with REST APIs; deployed on Linux/Nginx handling 50+ concurrent users at <200ms response.

EDUCATION

B.E. Computer Science & Engineering — Dr Ambedkar Institute of Technology, Bengaluru Jan 2020 – Jun 2024 | CGPA: 8.2/10